

Homework — 2.2.2 Computational Methods

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 2.2.2 lesson. Show traces/working.

Part A — This week's topic (~14 marks)

1. Complete the table by naming the **computational method** being used in each scenario. [2] (AO1/AO2)

Scenario	Computational method
A weather service displays thousands of temperature readings as a colour-coded heat map instead of a long table of numbers.	
A supermarket analyses millions of loyalty-card transaction records to discover unexpected patterns and relationships in shopping habits.	
A delivery firm plans van routes with a rule of thumb that quickly produces a good-enough route rather than the guaranteed best one.	
Before building a new ticketing website, engineers model how the proposed design will cope with the expected peak load.	

2. Write the letter of the correct definition in the Match column for each term. Two definitions are not needed. [2] (AO1)

Term	Match
Decomposition	
Divide and conquer	
Abstraction	
Problem recognition	

Letter	Definition
A	Removing or hiding unnecessary detail to focus on what matters for the problem
B	Identifying and understanding a problem — its inputs, outputs and requirements — before attempting to solve it
C	Breaking a problem into smaller, more manageable sub-problems, which may each be different kinds of task
D	Repeatedly splitting a problem into smaller instances of the same problem, solving these, and combining the results where needed
E	A rule of thumb that finds a good-enough answer quickly without guaranteeing the best one
F	Searching very large data sets to find unexpected patterns and relationships

3. A photo-sharing website processes every uploaded photo in three stages — resize, apply filter, add watermark — with each stage handled by a separate processor. Photos P1, P2 and P3 arrive one after

another.

(a) Complete the table to show which photo each stage is working on at each time step when the stages are **pipelined**. Use "—" for an idle stage. Time step 1 has been done for you. **[2]** (AO2)

Time step	Resize	Apply filter	Add watermark
1	P1	—	—
2			
3			
4			
5			

(b) Using the table, explain why pipelining increases the **throughput** of this system. **[1]** (AO2)

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4. A robot explores a maze. It follows a path until it reaches a dead end, then returns to the most recent junction and tries a different path, repeating this until it finds the exit.

(a) State the name of the problem-solving technique the robot is using. **[1]** (AO1)

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(b) The five stages of the technique are listed below out of order. Number them 1–5. **[2]** (AO2)

Stage	Order (1–5)
Reach a dead end where no forward move is possible.	
Choose a path at the first junction and start following it.	
Try a different, untried path from that junction, continuing the exploration.	
Continue extending the path, junction by junction, while forward moves are available.	
Undo the most recent choice and return to the last junction that still has an untried option.	

5. A delivery firm uses a **heuristic** to plan the routes for its vans rather than an exact method.

(a) State what is meant by a **heuristic**. **[1]** (AO1)

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(b) Explain **why** a heuristic, rather than an exact algorithm, is appropriate for a route-planning problem such as the Travelling Salesman Problem. **[2]** (AO2)

6. State one feature of a problem that makes it possible to solve computationally. [1] (AO1)

Part B — Retrieval: keep it fresh (~6 marks)

7. (2.2.1) State the defining difference between a **function** and a **procedure**. [1]

8. (2.2.1) A recursive subroutine has no reachable **base case**. State the run-time error that results and explain why it occurs. [2]

9. (2.1.2) State what is meant by **thinking ahead**, and give **two** examples of where it is used when designing a system. [3]